1. Note: this are the same matrices from 5.1 exercise 1.

a. Oops! I did this one in class. $\det(A - tI_2) = \det\begin{bmatrix} 1-t & 5 \\ 2 & 4-t \end{bmatrix} = (1-t)(4-t) - 10 = t^2 - 5t - 6 = (t-6)(t+1) = 0$ when $t = 6, -1$ so A has eigenvalues $\lambda = -1, 6$.

$E(-1) : N(A + I_2) = N\left(\begin{bmatrix} 1+1 & 5 \\ 2 & 4+1 \end{bmatrix}\right) = N\left(\begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}\right)$ and $\begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$ so $2x_1 = -5x_2 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ gives a basis for $E(-1)$.

$E(6) : N(A - 6I_2) = N\left(\begin{bmatrix} 1-6 & 5 \\ 2 & 4-6 \end{bmatrix}\right) = N\left(\begin{bmatrix} -5 & 5 \\ 2 & -2 \end{bmatrix}\right)$ and $\begin{bmatrix} -5 & 5 \\ 2 & -2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ so $x_1 = x_2 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ gives a basis for $E(6)$.

b. $\det(A - tI_2) = \det\begin{bmatrix} 0-t & 1 \\ 1 & 0-t \end{bmatrix} = (-t)(-t) - 1 = t^2 - 1 = (t-1)(t+1) = 0$ when $t = 1, -1$ so A has eigenvalues $\lambda = -1, 1$.

$E(-1) : N(A + I_2) = N\left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right)$ and $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ so $x_1 = -x_2 = 0 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ gives a basis for $E(-1)$.

$E(1) : N(A - I_2) = N\left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right)$ and $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ so $x_1 = x_2 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ gives a basis for $E(1)$.

c. $\det(A - tI_2) = \det\begin{bmatrix} 10-t & -6 \\ 18 & -11-t \end{bmatrix} = (10-t)(-11-t) + 108 = t^2 + t - 2 = (t-1)(t+2) = 0$ when $t = 1, -2$ so A has eigenvalues $\lambda = -2, 1$.

$E(-2) : N(A + 2I_2) = N\left(\begin{bmatrix} 12 & -6 \\ 18 & -9 \end{bmatrix}\right)$ and $\begin{bmatrix} 12 & -6 \\ 18 & -9 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1/2 \\ 0 & 0 \end{bmatrix}$ so $x_1 = 1/2 x_2 = 0 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}$ gives a basis for $E(-2)$.

$E(1) : N(A - I_2) = N\left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right)$ and $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ so $x_1 = x_2 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ gives a basis for $E(1)$.

j. $\det(A - tI_3) = \det\begin{bmatrix} 2-t & 0 & 1 \\ 0 & 1-t & 2 \\ 0 & 0 & 1-t \end{bmatrix} = (1-t)\det\left(\begin{bmatrix} 2-t & 1 \\ 0 & 1-t \end{bmatrix}\right) = (1-t)(2-t)(1-t) = 0$ when $t = 1, 2$ so A has eigenvalues $\lambda = 1, 2$.

$E(1) : N(A - I_3) = N\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}\right)$ and $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ so $x_1 = x_3 = 0 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$E(2) : N(A - 2I_3) = N\left(\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix}\right)$ and $\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ so $x_2 = x_3 = 0 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

m. $\det(A - tI_3) = \det\begin{bmatrix} 1-t & -1 & 4 \\ 3 & 2-t & -1 \\ 2 & 1 & -1-t \end{bmatrix} = (t-1)(t+2)(t-3) = 0$ when $t = 1, -2, 3$ so A has eigenvalues $\lambda = 1, -2, 3$ (done in class).

$E(-2) : N(A + 2I_3) = N\left(\begin{bmatrix} 3 & -1 & 4 \\ 3 & 4 & -1 \\ 2 & 1 & 1 \end{bmatrix}\right)$ and $\begin{bmatrix} 3 & -1 & 4 \\ 3 & 4 & -1 \\ 2 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ so

$x_1 = -x_3$ and $x_2 = x_3 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$E(1) : N(A - I_3) = N\left(\begin{bmatrix} 0 & -1 & 4 \\ 3 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}\right)$ and $\begin{bmatrix} 0 & -1 & 4 \\ 3 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix}$ so

$x_1 = -x_3$ and $x_2 = 4x_3 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$

$E(3) : N(A - 3I_3) = N\left(\begin{bmatrix} -2 & -1 & 4 \\ 3 & -1 & -1 \\ 2 & 1 & -4 \end{bmatrix}\right)$ and $\begin{bmatrix} -2 & -1 & 4 \\ 3 & -1 & -1 \\ 2 & 1 & -4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$ so

$x_2 = x_3$ and $x_2 = 2x_3 \Rightarrow \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$